Security Operations Center (SOC)

Incident Prioritization, Classification and Response – Cyber Security Practical Report

1. Introduction

A Security Operations Center (SOC) monitors and protects an organization's infrastructure from cyber threats. SOC analysts review logs, detect suspicious activity, investigate incidents, and respond quickly to reduce damage. Modern SOC environments use SIEM platforms, threat intelligence feeds, and automated response tools.

2. Incident Prioritization

Priority	Description	Example
Critical	Major business impact	Ransomware encrypting servers
High	Unauthorized access	Admin account compromise
Medium	Suspicious activity	Multiple failed logins
Low	Minor alerts	Network scanning

Vulnerabilities are also prioritized using the CVSS scoring system. A CVSS score above 9 is considered critical and requires immediate remediation.

3. Incident Classification

Category	Description
Malware	Malicious software infection
Phishing	Fraudulent email attack
DDoS	Overloading servers with traffic
Insider Threat	Malicious internal activity
Data Exfiltration	Unauthorized data transfer

Frameworks such as MITRE ATT&CK; and VERIS help analysts map incidents to attacker techniques and improve threat detection.

4. Basic Incident Response

Phase	Description
Preparation	Develop policies and tools
Identification	Detect suspicious activity
Containment	Isolate compromised systems
Eradication	Remove malware
Recovery	Restore normal operations
Lessons Learned	Improve defenses

5. SOC Workflow

Logs	→	SIEM	→	Alert Analysis	→	Incident Response	→	Recovery

6. Incident Response Lifecycle

Preparation	→	Identification	→	Containment	→	Eradication	→	Recovery

7. Conclusion

SOC teams detect, analyze, and respond to cybersecurity threats. Proper prioritization, classification, and structured incident response procedures enable organizations to quickly mitigate attacks and improve their security posture.

SOC Practical Report – Alert Management, CVSS Analysis and Incident Response

This report demonstrates practical Security Operations Center (SOC) activities including alert monitoring, CVSS vulnerability prioritization, SIEM dashboard analysis, MITRE ATT&CK; mapping, and incident response workflow. The screenshots below represent real-world cybersecurity monitoring dashboards and vulnerability management systems.

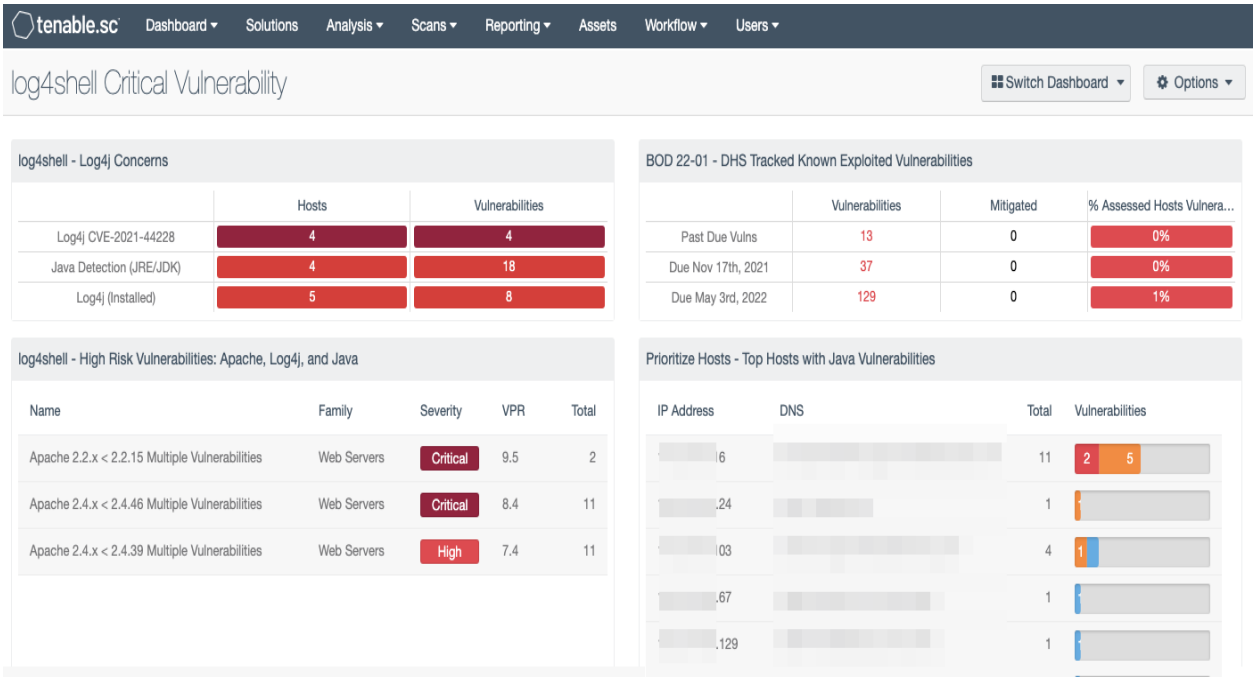
1. CVSS Vulnerability Prioritization

The Common Vulnerability Scoring System (CVSS) is used to measure the severity of vulnerabilities. Security teams use CVSS exploitability metrics and remediation levels to determine which vulnerabilities require immediate patching.

Analysis: The dashboard shows exploitability and remediation heat maps for vulnerabilities. Red sections indicate high risk vulnerabilities that currently lack remediation solutions, while green sections represent vulnerabilities that already have official fixes available.

2. Log4Shell Critical Vulnerability Monitoring

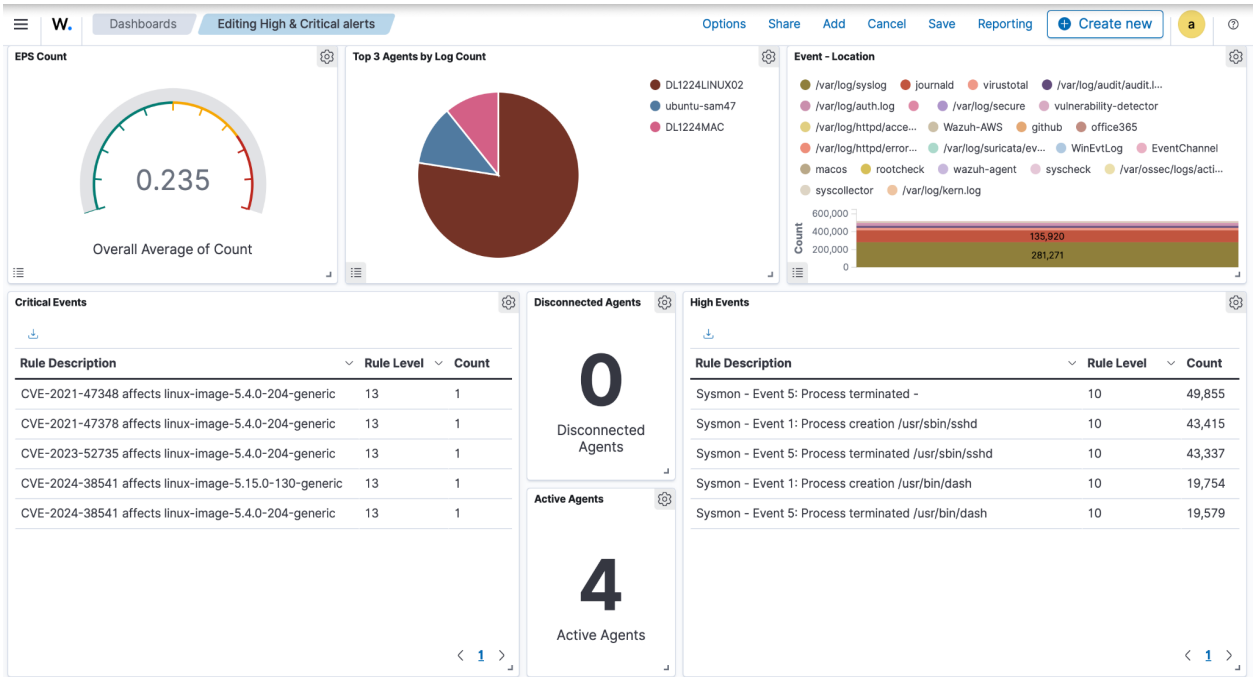
The Log4Shell vulnerability (CVE-2021-44228) is a critical remote code execution vulnerability in Apache Log4j. Security teams use vulnerability scanners and dashboards to track affected hosts and prioritize remediation.



Result: The dashboard shows multiple hosts affected by Log4j vulnerabilities. Several Apache server versions were identified as critical severity with high VPR scores, indicating an urgent need for patching or mitigation.

3. SIEM Monitoring Dashboard (Wazuh)

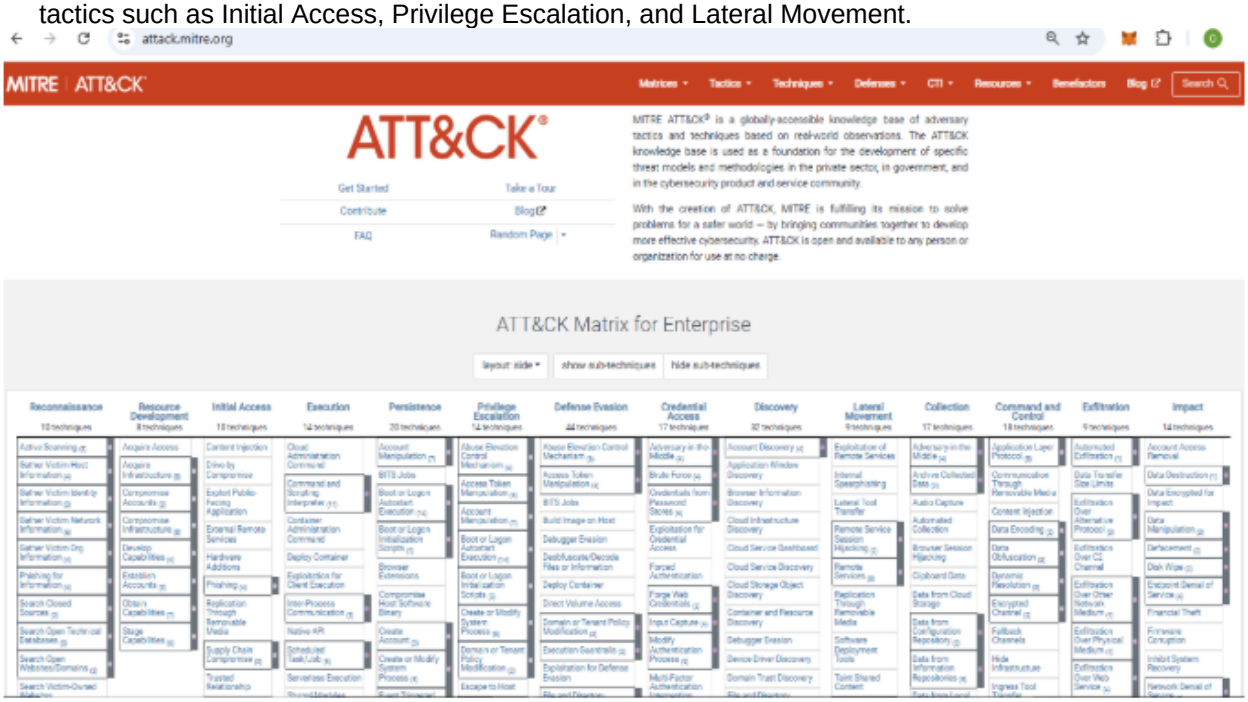
A SIEM platform aggregates logs from multiple systems and visualizes security events in real time. The dashboard below displays alert statistics, active agents, event counts, and rule-based detections.



Analysis: The dashboard indicates several high event counts from system monitoring tools such as Sysmon. The presence of multiple critical event detections suggests continuous monitoring of system processes and security logs.

4. MITRE ATT&CK; Framework Mapping

The MITRE ATT&CK; framework is widely used in SOC operations to classify attacker techniques and understand adversary behavior. It helps analysts map alerts and incidents to specific attack tactics such as Initial Access, Privilege Escalation, and Lateral Movement.



Analysis: Using the MITRE ATT&CK; matrix allows analysts to map alerts like phishing or brute-force login attempts to known attacker techniques. This helps security teams improve detection rules and strengthen defensive controls.

5. Incident Response Timeline

Timestamp	Action
2025-08-18 14:00	Suspicious activity detected
2025-08-18 14:10	Endpoint isolated from network
2025-08-18 14:30	Malware indicators analyzed
2025-08-18 15:00	Threat intelligence verification completed
2025-08-18 15:30	Incident escalated to Tier-2 SOC team

Conclusion: This SOC practical demonstrates how security analysts prioritize vulnerabilities using CVSS, monitor security events through SIEM dashboards, map incidents using MITRE ATT&CK;, and follow structured incident response procedures to mitigate cyber threats.

SOC Practical Report – Evidence Preservation and Capstone Attack Response

This practical demonstrates digital evidence preservation and a full SOC alert-to-response cycle including attack simulation, detection, triage, containment, and reporting.

1. Volatile Data Collection using Velociraptor

Velociraptor - Network Connections Collection

Query: SELECT * FROM netstat

Host: Windows-VM01

Active Connections Detected: 14

Export Format: CSV (netstat_results.csv)

Analysis: Velociraptor was used to collect volatile network connection data from a Windows virtual machine using the query SELECT * FROM netstat. The results were exported to CSV format for forensic investigation.

2. Memory Dump Acquisition

Velociraptor - Memory Acquisition

Artifact: Artifact.Windows.Memory.Acquisition

Target: Server-X

Memory Dump File: serverx_memory_dump.mem

Collection Status: Completed

A full memory dump of Server-X was collected to capture running processes, network connections, and malicious artifacts present in RAM.

3. Evidence Hash Verification

FTK Imager - Evidence Hash Verification

Evidence File: serverx_memory_dump.mem

Hash Algorithm: SHA256

Computed Hash: 8fa14cdd754f91cc6554c9e71929cce7...

Verification Status: Verified

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X Dump	SOC Analyst	2025-08-18	SHA256: 8fa14cdd754f91cc6554c9e

4. Attack Simulation using Metasploit

Metasploit - Attack Simulation

Module: exploit/unix/ftp/vsftpd_234_backdoor

Target: Metasploitable2 (192.168.1.100)

Exploit Status: Session Opened

Result: Remote shell access gained

The Metasploit framework was used to exploit the VSFTPD backdoor vulnerability on a Metasploitable2 system. A remote shell session was successfully established demonstrating unauthorized access.

5. Detection and Alert Triage (Wazuh)

Wazuh SIEM Alert

Alert: VSFTPD Backdoor Exploit Detected

Source IP: 192.168.1.100

Rule Level: High

Mapped MITRE Technique: T1190

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 11:00:00	192.168.1.100	VSFTPD exploit	T1190

6. Response and Containment using CrowdSec

CrowdSec Response

Action: Block Attacker IP

Blocked IP: 192.168.1.100

Firewall Action: Ban Applied

Verification: Ping test blocked

The attacker IP was blocked using CrowdSec firewall remediation. Network connectivity tests confirmed that the malicious IP address could no longer communicate with the protected system.

7. Executive Incident Report

During the simulation, a VSFTPD backdoor vulnerability was exploited on a vulnerable Linux server. The attack was detected by the SIEM platform which generated an alert indicating suspicious FTP activity from the attacker IP address. SOC analysts investigated the alert and confirmed the exploit attempt. The affected host was isolated and forensic evidence including memory dumps and network connection logs were collected for analysis. The attacker IP address was blocked using CrowdSec firewall controls. After containment, the vulnerability was patched and security monitoring rules were updated to prevent similar attacks.

8. Stakeholder Briefing

A security incident involving unauthorized access to a test server was detected and contained. Our monitoring system quickly identified suspicious activity and the security team isolated the affected system to prevent further impact. The attacker’s IP address was blocked and the vulnerability responsible for the incident was patched. No sensitive data was compromised and additional monitoring controls have been implemented to strengthen security defenses.